전기 화재 단락흔 분류를 위한 3가지 경량 딥러닝 모델의 적용과 비교 분석

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Application and Comparative Analysis of Three Lightweight Deep Learning Models for Electric Fire Short-Circuit Trace Classification

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Abstract - Electrical short circuits are a major cause of electrical fires, making accurate identification of short-circuit traces important in fire investigations. This paper evaluates three lightweight deep learning models MobileNetV2, MobileNetV3, and EfficientNet for classifying primary short-circuit traces (PSCT) and secondary short-circuit traces (SSCT). A dataset of 752 images was used with an 80% - 20% training - testing split, applying data augmentation and transfer learning. Results show that EfficientNet achieved the highest validation accuracy (96.05%), while MobileNetV3 produced the lowest validation loss, demonstrating the effectiveness of lightweight CNN models for automated fire investigation.

1. Introduction

Electrical fires threaten property and human safety, maki ng it important to identify their origin. During fires, electri cal conductors may develop short-circuit traces, which are melting marks formed by abnormal current and heat. Thes e traces are classified as Primary Short-Circuit Traces (P SCT), which occur before fire ignition, and Secondary Sho rt-Circuit Traces (SSCT), which occur after the fire begin s.

Traditional investigation methods rely on visual inspection and metallurgical analysis, which can be time-consuming and require expert interpretation. Recent advances in deep learning, particularly convolutional neural networks (CNNs), enable automated image classification. Lightweight architectures such as MobileNetV2, MobileNetV3, and EfficientNetare especially suitable because they require fewer computational resources.

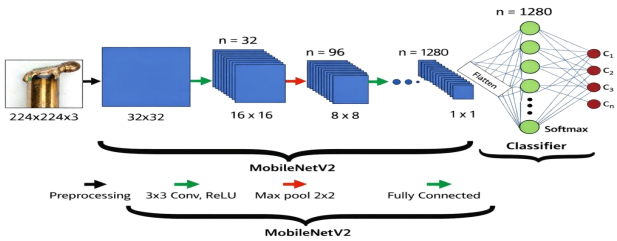
This paper evaluates these three models for classifying electric fire short-circuit traces and explores their potential use in practical fire investigation systems, including mobile applications.

2. Related Work

Deep learning is widely used for image classification, with convolutional neural networks (CNNs)effectively learning hierarchical visual features. Early models such as AlexNet improved large scale image recognition, while later architectures like VGGNet and ResNet increased accuracy through deeper networks and residual learning.

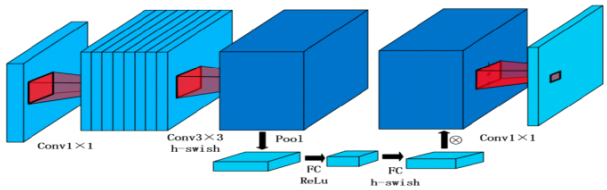
However, these models require high computational resources. To address this, lightweight CNN architectures were developed for efficient performance on mobile and embedded systems. In this paper, the following lightweight CNN models are used:

- MobileNetV2: introduced depthwise separable convolutions and inverted residual blocks to reduce computational complexity while maintaining strong feature extraction capability [1],[4].



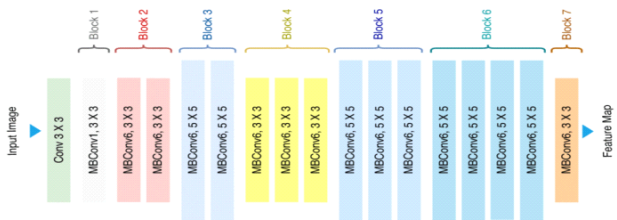
<Figure 1> MobileNetV2 network architecture

- MobileNetV3: improved upon MobileNetV2 by incorporating squeeze and excitation attention mechanisms and optimized activation functions through neural architecture search [2],[5].



<Figure 2> Block structure of MobileNetV3

- EfficientNet: introduced a compound scaling method that simultaneously scales network depth, width, and resolution, enabling improved accuracy with fewer parameters [3],[6].



<Figure 3> Architecture of EfficientNet

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3. Proposed Approach

3.1. Data preparation and preprocessing

The dataset used in this paper consists of 752 images of electric short-circuit traces collected from electrical conductor samples under controlled experimental conditions.
The images represent two categories of short-circuit traces commonly observed during fire investigations: Primary Short-Circuit Traces (PSCT) and Secondary Short-Circuit Traces (SSCT) All images were resized to 224 × 224 pixels to match the input requirements of the deep learning models.

3.2 Data augmentation

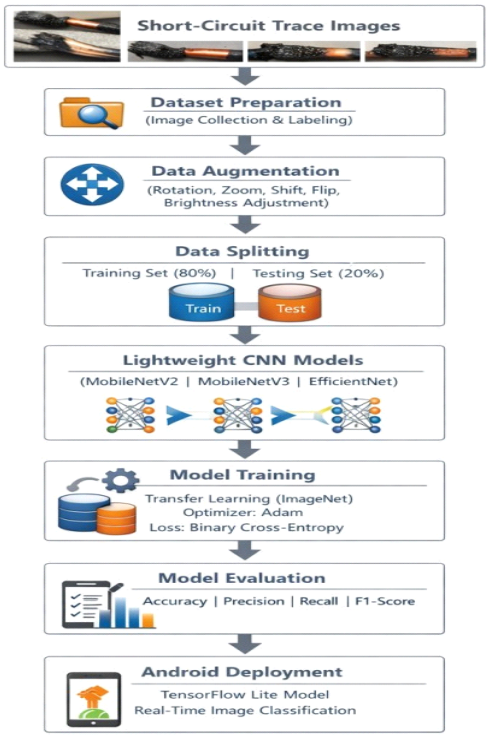
To improve model generalization and reduce over-fitting, several data augmentation techniques were applied using the TensorFlow Keras ImageDataGenerator. The following augmentation techniques were used:

- Rotation within ±20°
- Width and height shifting up to 10%
- Shear transformation (0.1)
- Zoom range ±10%
- Horizontal flipping

These transformations increase dataset diversity and allow the model to learn robust visual representations.

3.3. Optimization algorithm

In this paper, the Adam optimizer (Adaptive Moment Estimation) was used for training the models. Adam is a widely used stochastic optimization algorithm that combines the advantages of two other popular optimization methods: AdaGrad and RMSProp.



<Figure 4> Proposed Framework for Electric Fire Short-Circuit Trace Classification

4. Result and Analysis of Experiment

Experiments in this paper were conducted on a 64-bit Windows 11 system with an AMD Ryzen 9 5900X processor and 32GB RAM.
The models were implemented using Python, TensorFlow, and Kers.

<Table 1> Model Performance

Model	Training Accuracy	Validation Accuracy	Validation Loss
MobileNetV2	91.39%	92.76%	0.1523
MobileNetV3	93.59%	95.39%	0.1044
EfficientNet	91.65%	96.05%	0.1739

EfficientNet achieved the highest validation accuracy.

5. Discussion

The results show that lightweight CNN architectures can effectively classify electric fire short-circuit traces while requiring fewer parameters and lower computational resources than traditional models, making them suitable for deployment on resource-constrained devices.
Among the evaluated models, EfficientNet achieved the highest validation accuracy (96.05%), indicating strong feature extraction capability for distinguishing between primary short-circuit traces (PSCT) and secondary short-circuit traces (SSCT). Its compound scaling method improves model performance while maintaining efficiency. MobileNetV3 demonstrated stable training performance and produced the lowest validation loss, suggesting good generalization ability. The squeeze-and-excitation mechanism allows the model to focus on important visual features in short-circuit trace images.

6. Conclusion

This paper evaluated three lightweight deep learning architectures MobileNetV2, MobileNetV3, and EfficientNet for classifying primary and secondary short-circuit traces in electrical fire investigations.
Experimental results showed that EfficientNet achieved the highest validation accuracy of 96.05%, while MobileNetV3 demonstrated stable training performance.
The results indicate that lightweight CNN architectures can effectively support automated fire investigation systems.
Future work will focus on expanding the dataset, improving classification performance, and developing a fully functional mobile-based fire investigation tool.

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